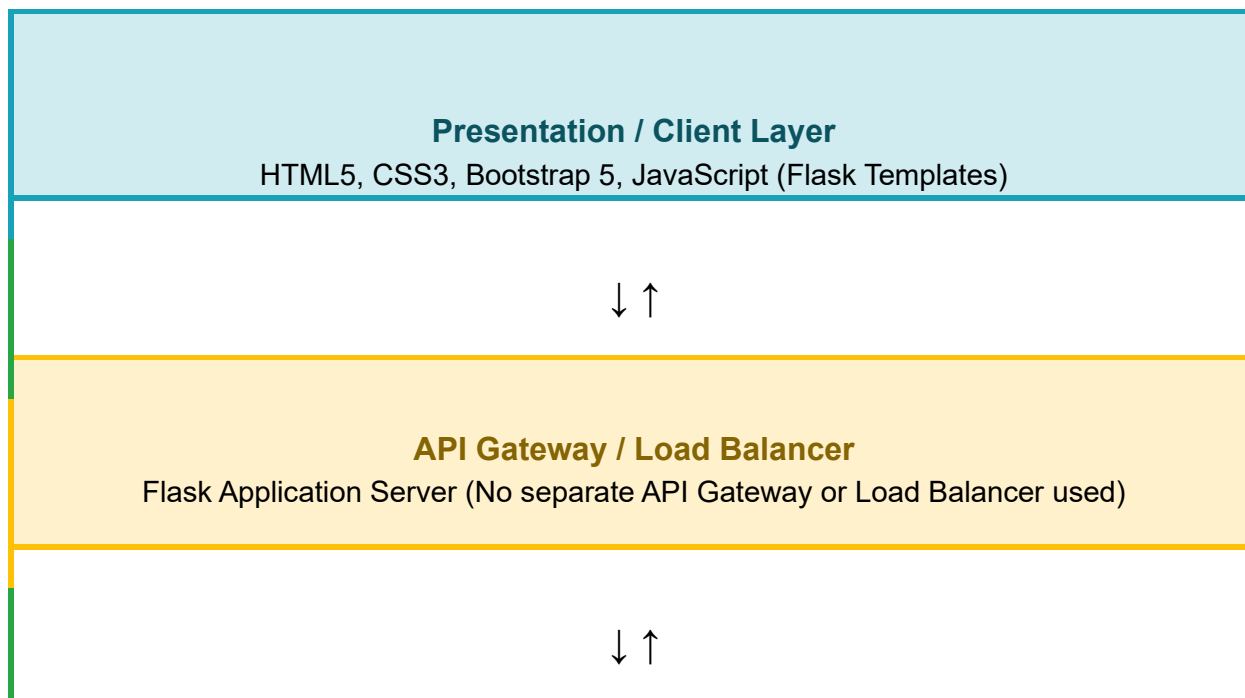


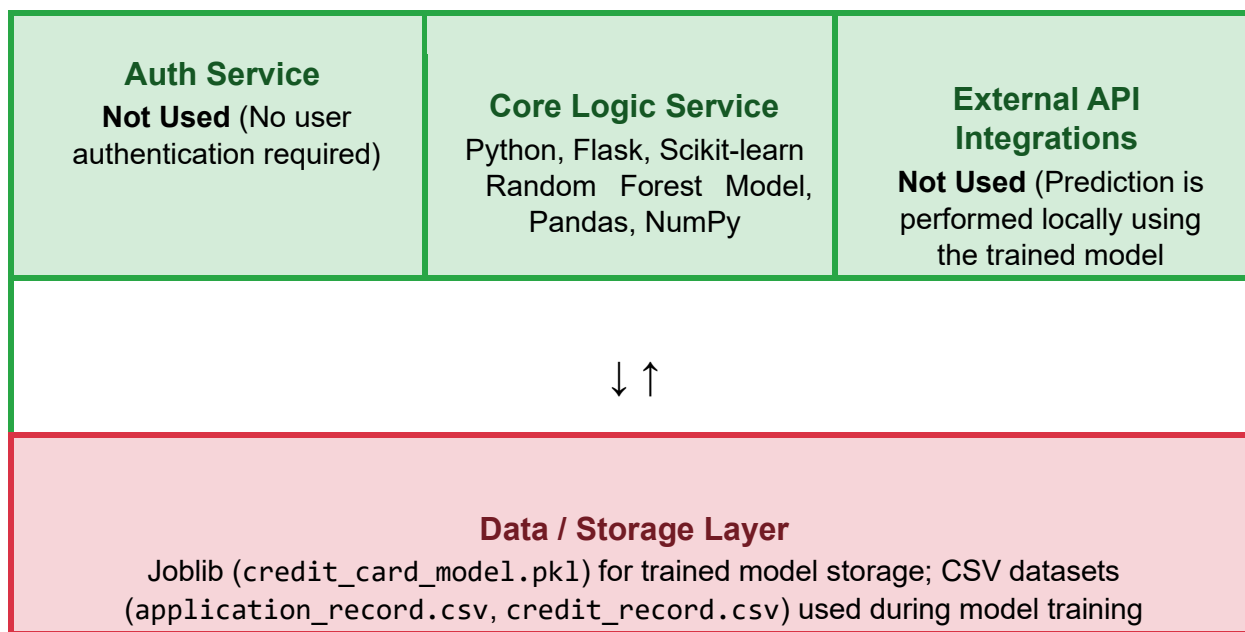
## Solution Architecture

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 15 July 2026                    |
| <b>Team ID</b>       | SWTID-2026-6771                 |
| <b>Project Name</b>  | Credit card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





## Component Description Table

| Component Name            | Description / Role in Architecture                                                                                                                              | Technologies Used                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Presentation Layer</b> | Provides a user-friendly interface for applicants to enter personal, financial, and employment details and displays the credit card approval prediction result. | HTML5, CSS3, Bootstrap 5, JavaScript, Flask Templates                                 |
| <b>API Gateway</b>        | Acts as the entry point for handling HTTP requests from the user interface and routes them to the Flask backend for processing.                                 | Flask Web Server                                                                      |
| <b>Microservices</b>      | Performs data preprocessing, loads the trained Random Forest model, processes applicant information, and generates the approval prediction.                     | Python, Flask, Scikit-learn, Pandas, NumPy, Joblib                                    |
| <b>Database / Storage</b> | Stores the trained machine learning model used for prediction. The datasets are used only during model training and are not accessed during prediction.         | Joblib (credit_card_model.pkl), CSV files (application_record.csv, credit_record.csv) |

